

Existing Problematics

Core Decision Problems:

- **Berth planning** — which berth should each arriving ship use?
- **Stowage planning** — where should each container be placed *inside* the ship?
- **Yard planning** — where should containers be stored in the terminal yard?
- **Container sequencing** — in what order should containers be loaded/unloaded?
- **Vehicle assignment** — which vehicle should transport which container?

Ship Planning / Waterside Operations

Operations around the **ship and quay** form the first critical layer. Every minute a vessel spends idle at a berth represents significant cost, making these decisions time-sensitive and high-stakes.

- **Berth allocation** — deciding where and when a ship should dock; balancing multiple vessels against limited berth space and time windows.
- **Stowage planning** — placing containers on the ship respecting weight limits, stability constraints, and the order of port calls (to avoid overstowage).
- **Quay-crane scheduling** — assigning cranes to containers, minimizing makespan while respecting crane interference and safety distances.
- **Double cycling** — coordinating cranes to load and unload in a single sweep, reducing total moves.

Storage and Stacking

The **container yard** is the terminal's buffer zone. Efficient yard management is central to minimizing re-handling — one of the largest hidden costs in port operations.

- **Yard allocation** — assigning containers to storage blocks; optimized via genetic algorithms and metaheuristics in the literature.
- **Container stacking** — deciding stack height and order so that containers needed next are always on top; minimizing reshuffle moves.
- **Re-handling / reshuffling** — when a needed container is buried, others must be moved; reducing this number is a key optimization target.
- **Yard-crane scheduling & automation** — coordinating rail-mounted or rubber-tired gantry cranes, including fully automated storage systems.

Horizontal Transportation

Moving containers **between the quay and the yard** ties the two previous layers together. The

primary resources are **Automated Guided Vehicles (AGVs)** or conventional trucks.

- **Dispatching** — which vehicle picks up which container, and when?
- **Routing** — which path does the vehicle take through the terminal?
- **Empty-trip reduction** — scheduling return trips to carry another container instead of traveling empty.
- **Fleet utilization** — balancing load across vehicles to avoid bottlenecks.

Research in this area relies heavily on simulation-based comparisons of routing strategies, with AGV systems receiving growing attention as terminals automate.

Data Availability

based on my research , existing works doesnt use standard source for evaluation and training , they generate **synthetic data** , one example is this : [generator.py]

